Analysis and Prediction of Consumer Financial Complaints

SIADS 699 Capstone Project

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PROBLEM

Use CFPB complaint data to understand what problems consumers describe, how complaints vary across geography and time, and whether complaint features help predict company-response outcomes.

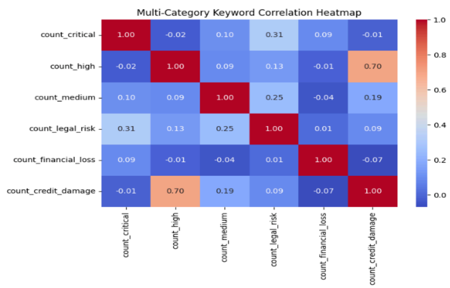
DATA

CFPB Consumer Complaint Database.
2025 narrative dataset: 1,221,970 complaints.
Historical sample for geography forecasting: 25% sample from 2011–2026.

METHODS

Natural language processing, unsupervised learning, geographic analysis, time-series forecasting, and supervised machine learning.

1. NLP and Severity Analysis



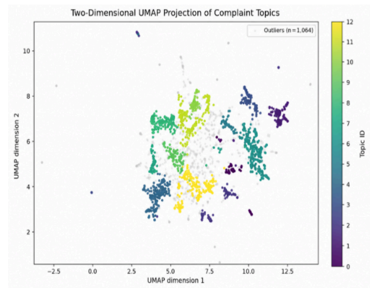
Severity Indicator Framework

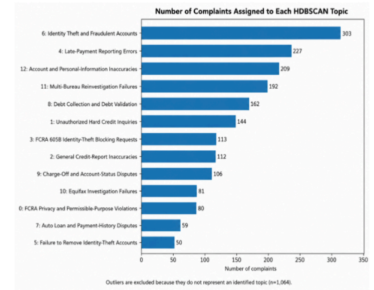
Category	Example Indicators
Critical Severity	foreclosures, bankruptcy, lawsuit, garnishment, eviction, mortgage
High Severity	fraud, identity theft, phishing, unauthorized transaction, account takeover
Medium Severity	incorrect reporting, billing error, credit score concern, overdraft fee
Additional Dimensions	legal risk, financial loss, credit damage

- VADER sentiment captured tone, but not financial severity.
- A domain-specific severity framework using supply phrase matching identified critical, high, and medium severity indicators.
- The severity framework was considered useful: it identified financial-risk patterns beyond general sentiment and stayed similar under alternative weighting schemes.

Takeaway: ***sentiment alone did not capture financial severity in complaint narratives.***

2. Unsupervised Topic Analysis

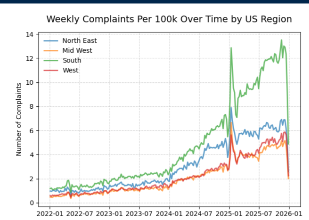


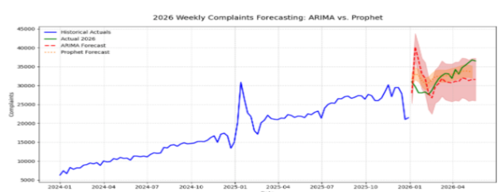


- Analyzed 2,902 filtered credit-reporting complaints.
- Sentence embeddings + UMAP + HDBSCAN identified 13 complaint topics.
- Largest topics included identity theft and fraudulent accounts, late-payment reporting errors, account and personal-information inaccuracies, and multi-bureau reinvestigation failures.

Takeaway: ***narratives revealed more specific themes than broad CFPB categories.***

3. Geographic and Forecasting Analysis





Metric	SARIMAX	Prophet
MAE	3,374.52	1,814.57
RMSE	4,227.78	2,209.33
MAPE	10.67%	5.93%
MASE	0.49	0.26

- After normalization, the South had the highest complaint rate.
- No explanation for volatility in early 2025 wasn't found in the dataset.
- All four regions showed similar seasonal patterns.
- Prophet outperformed SARIMAX on the 2026 holdout.

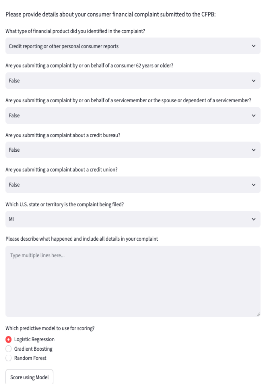
Takeaway: ***complaint activity varied by region and time, and Prophet gave the best short-term forecast in the 2026 holdout.***

4. Supervised Learning

Model Comparison

Model	Average Precision
Logistic Regression	0.81
Gradient Boosting	0.82
Random Forest	0.82

Streamlit App Interface



- Target: company-response outcome.
- Compared logistic regression, gradient boosting, and random forest.
- Average Precision was the main comparison metric because the classes were imbalanced.
- Logistic regression remained useful because the small performance gap made interpretability valuable.
- The models were also deployed in a Streamlit app.

Takeaway: ***complaint text and structured complaint features helped predict company-response outcomes.***